

ORF 245: Fundamentals of Statistics – Spring 2025

Precept 4

1. (Joint Distribution)

- (a) Joint PMF $p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$ and Joint PDF $f_{X,Y}(x, y)$ (function satisfying $\mathbb{P}((X, Y) \in A) = \int_A f_{X,Y}(x, y) dx dy$.)
- (b) Marginal PMF $p_X(x) = \sum_{y \in \text{Range } Y} p(x, y)$ and Marginal PDF $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$
- (c) Joint can determine Marginal counterpart while the converse is false
 - $X \sim \text{Ber}(1/2)$, $Y_1 \sim \text{Ber}(1/2)$ indep of X , $Y_2 = X$: marginal same, joint different
 - The converse is true if they are independent.
- (d) Conditional PMF/PDF
- (e) Independence: X and Y are independent if $p_{X,Y}(x, y) = p_X(x)p_Y(y)$ (for discrete R.V.) and $f_{X,Y}(x, y) = f_X(x)f_Y(y)$ (for continuous R.V.)
 - That also holds for CDF.
- (f) Expectation: $\mathbb{E}[h(X, Y)] = \sum_{x,y} h(x, y)p_{X,Y}(x, y)$; $\mathbb{E}[h(X, Y)] = \int h(x, y)f_{X,Y}(x, y) dx dy$
 - Linearity of Expectation: $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.
 - Covariance: $\mathbb{C}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.
 - Independence $\rightarrow$ Covariance = 0 (uncorrelated), the converse is not true (but is true for joint Gaussian random variable)
 - $\mathbb{V}(X + Y) = \mathbb{V}(X) + \mathbb{V}(Y) + 2\mathbb{C}(X, Y)$: uncorrelated, further have $\mathbb{V}(X + Y) = \mathbb{V}(X) + \mathbb{V}(Y)$.
- (g) Correlation Coefficient: $\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sigma_X \cdot \sigma_Y} = \frac{\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]}{\sqrt{\mathbb{E}[(X - \mathbb{E}[X])^2] \cdot \mathbb{E}[(Y - \mathbb{E}[Y])^2]}}$
 $\rho_{XY} \in [-1, 1]$. A value of +1 implies perfect positive correlation (linear relationship), and a value of -1 implies perfect negative correlation (linear relationship). A value of 0 implies no correlation.

2. (A uniform distribution over a square region) Let (X, Y) be uniformly distributed in the region $\{(x, y) | |x| + |y| < 1\}$.

- What is the PDF of (X, Y) ?
- What is the covariance of X and Y ?
- What are the expectations of X and Y ? What is the conditional PDF of X given $Y = 1/2$?

3. (Linearity of Expectation) Calculate the expected number of cards that need to be turned over in a regular 52-card deck to see the first ace.

4. (Q&A)